Problem 3: Simulation vs. Practical Development

Technical proposal from the lead designer

1. My decision in one paragraph

I would not pick only one side. My proposal is a **simulation-guided, prototype-first hybrid**: use cheap, fast simulation in week 1 to size the aircraft and avoid obvious mistakes, then build a real prototype as early as possible (end of week 2), and let flight tests decide the final design. Simulation is good at telling us *which direction* to go; only a real aircraft tells us *whether we got there*. With only four weeks and a small budget, we cannot afford to simulate for three weeks and discover a problem on the runway, and we cannot afford to build blindly and crash our only airframe either.

Assumptions (all assumed, for this proposal). Team of four, four weeks, budget about INR 8,000, one competition aircraft with a spare set of foam/balsa parts. Mission: take off from a short runway (about 10 m), fly a circuit and carry a small payload. Starting design: mass 1.5 kg, wing area 0.30 m^2, electric motor and propeller.

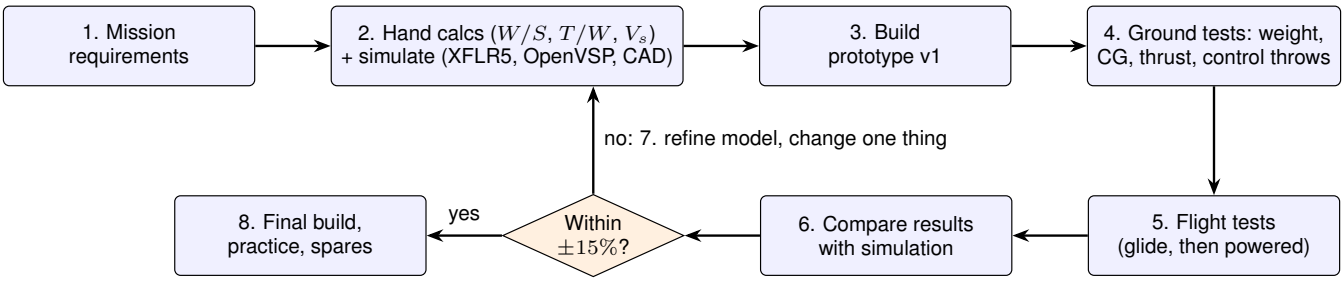
2. Advantages and limitations

	Simulation-led (XFLR5, OpenVSP, CAD, ANSYS)	Prototype-led (build, fly, fix)
Advantages	Cheap to repeat; many variants in hours; shows trends (airfoil, stability, drag); no crash risk; good for sizing before buying parts.	Shows real behaviour: weight, build quality, vibration, control feel, battery performance; finds problems models never include.
Limitations	Only as good as its inputs and assumptions. At low Reynolds numbers ($\sim 10^5$) XFOIL is noisy near stall; it ignores build errors, warped wings and wiring problems. High-end tools (ANSYS) cost days of learning.	Each iteration costs materials and days; crashes destroy parts; hard to measure small effects; risk of random changes without understanding why.

3. Comparison table

Criterion	Simulation-led	Prototype-led	Hybrid (my choice)
Cost	Low (software is free or student-licensed)	Medium–high (each rebuild)	Low–medium: one prototype and one spare
Time	Slow to start (learning, setup)	Fast first flight, slow to converge	Balanced: simulate week 1, build by week 2
Accuracy	Good for trends, poor for absolute values	Best, but noisy (wind, pilot)	Good: model is corrected using measurements
Iteration speed	Very fast	Slow (hours to days)	Fast on paper, slow only for key changes
Failure detection	Finds design errors, not build errors	Finds real failures, often late and expensively	Finds both, earlier
Final performance	May look good but not fly as predicted	Good if time allows, can be sub-optimal	Best chance within four weeks

4. Proposed workflow



Steps 1–2 happen in week 1, steps 3–4 in week 2, steps 5–7 repeat in weeks 3–4 (at most two loops), and step 8 is the last few days. Each loop changes only one thing at a time, so we know what caused any improvement.

5. What to simulate and what to check physically

Simulate	Verify physically
Airfoil and wing: $C_{L,\max}$, C_m, drag polar (XFLR5/XFOIL)	**Actual mass and CG:** weigh every part, balance on the real aircraft
Stability: neutral point and static margin (OpenVSP/XFLR5, target about 5–15% of the chord)	**Static thrust and battery under load:** spring balance or luggage scale, voltage sag
Performance estimate: V_s, glide ratio, power needed for cruise (propulsion calculator such as eCalc)	**Structure and build quality:** wing tip load test, joint strength, warps, surface finish
Packaging and geometry: CAD fit of battery, servos, payload (Fusion 360)	**Controls and take-off:** control direction, throw, response, runway length needed

6. How to validate simulations with measurements

1. **Weigh and balance first.** Put the *measured* mass and CG back into the model. Most disagreements start here.
2. **Compare one number at a time:** predicted vs. measured stall speed, glide ratio (hand-launch glide over a measured distance and height loss), static thrust, and take-off distance. Record them in a table with the percent error.
3. **Set a tolerance.** I would accept about $\pm$15% on performance numbers at this Reynolds number. Anything larger means the model or the build is wrong.
4. **Fix the model, not the data.** If the aircraft glides worse than predicted, increase the assumed zero-lift drag in the model until it matches, then trust the model more for the next change.
5. **Log the conditions** (wind, battery, date) so a bad flight is explained and not just discarded.

7. The prototype cannot take off: what could be wrong?

Why a small mismatch is serious. Taking my starting design: $W = 14.7$ N, $S = 0.30$ m^2, $C_{L,\max} \approx 1.3$ (3D, assumed). Then $V_s = \sqrt{2W/(\rho S C_{L,\max})} \approx 7.8$ m/s and lift-off speed $\approx 1.2 V_s = 9.4$ m/s. With an average net thrust of about 9 N the ground roll $s = V^2/2a$ is about **8.5 m**. If the real aircraft weighs 1.9 kg and the thrust under load is only 7 N, the roll becomes about **19 m**, more than twice as long. Nothing is "broken", but it can no longer take off from a 10 m runway. (Numbers are assumed, for illustration.)

Plausible causes, grouped:

- **Weight:** the aircraft is heavier than modelled (glue, wiring, battery, payload).
- **Thrust:** propeller too large or too small for the motor, battery sag, low-quality ESC or battery, thrust measured on the bench was static and not at speed, motor or propeller direction reversed.
- **Aerodynamics:** wing incidence too low, warped wing, rough surface, less C_L than the 2D simulation, elevator trim or CG so far forward or aft that the nose will not rotate.
- **Ground and setup:** grass or soft ground, wheels dragging or misaligned, control surfaces moving the wrong way, runway too short or wind from behind.

Diagnostic plan (cheapest and quickest first, change one thing per step):

1. **Safety and basic checks:** control directions, motor and propeller direction, battery voltage *under load*.
2. **Weigh and balance:** compare mass and CG with the model; fix the CG first.
3. **Static thrust test:** compare thrust with the T/W needed (about 0.4–0.5 for take-off here). Try a different propeller if needed.
4. **Calculate:** use the measured mass and thrust in the take-off formula above to see the roll length you should expect.
5. **Hand-launch glide test:** if it glides well, the wing and trim are fine and the problem is power or ground; if it will not glide, the problem is aerodynamics or CG.
6. **Ground-run test:** test on smooth, hard ground; check wheels and rolling friction, add a slightly higher wing incidence if needed.
7. **Fix, retest, record:** write down each change and its result, then update the model (Section 6).

8. Bonus: four-week plan with contingency

Week	Work	Contingency
1	Requirements, hand calculations, XFLR5/OpenVSP studies, CAD, order parts	1 day for late parts or simulation problems
2	Build prototype v1, ground tests (weight, CG, thrust, controls)	2 days for rebuilds and repairs
3	Glide and powered flight tests, validation, refine, build v2 if needed	1 day for weather and a crash
4	Final tuning, practice flights, spare parts and packing	3 days kept free before the competition

About 20% of the time is kept as buffer. If week 2 slips, I cut optional work (extra simulations, v2 refinements), never the flight tests. Build one spare wing and one spare tail early, because those are the parts that break most often.

9. Why this approach

Simulation answers “what should we try first?” in hours at almost no cost; the prototype answers “does it actually work?” in days. Using each for what it is good at, and using measurements to correct the model, gives the best chance of a reliable aircraft within four weeks. The aircraft must fly in the competition, not in the software.

References

[1] Raymer, D.P., *Aircraft Design: A Conceptual Approach*, AIAA Education Series.
[2] Gudmundsson, S., *General Aviation Aircraft Design: Applied Methods and Procedures*, Butterworth-Heinemann, 2014.
[3] Anderson, J.D., *Fundamentals of Aerodynamics*, McGraw-Hill.
[4] Drela, M., “XFOIL: An Analysis and Design System for Low Reynolds Number Airfoils”, in *Low Reynolds Number Aerodynamics*, Springer, 1989.
[5] XFLR5 documentation and guidelines, `xflr5.tech`; OpenVSP (NASA), `openvsp.org`.
[6] Selig, M.S. et al., *Summary of Low-Speed Airfoil Data*, SoarTech, 1995–1997 (UIUC Low-Speed Airfoil Tests).